Theorem 2.10. *If the $n \times n$ matrix A is strictly diagonally dominant, then:*

1. *A is a non-singular matrix.*
2. *For every vector b and every starting guess, the Jacobi method applied to $Ax = b$ converges to the (unique) solution.*

Proof. Let $A = D + L + U$, where D is the diagonal part of A, and L and U are the strictly lower and upper triangular parts, respectively. Let $R = L + U$ denote the non-diagonal part of the matrix A. Note that the diagonal entries of R are zero.

To guarantee the convergence of the Jacobi method, we must show that the spectral radius of the iteration matrix is less than one, that, is $\rho(D^{-1}R) < 1$. In other words, we have to show that every eigenvalue of this matrix has magnitude strictly less than one.

Let λ be an arbitrary eigenvalue of $D^{-1}R$ with corresponding eigenvector v. By definition, we have:

$$D^{-1}Rv = \lambda v \implies Rv = \lambda Dv$$

Choose the eigenvector v such that its infinity norm is 1, i.e., $\|v\|_\infty = 1$. This means there exists an index m $(1 \leq m \leq n)$ such that the component $|v_m| = 1$, and for all other components, $|v_j| \leq 1$.

Considering the m-th component of the vector equation $Rv = \lambda Dv$, and recalling that $r_{mm} = 0$ (because, as already stated, all diagonal values of R are zero), we get:

$$\sum_{j \neq m}^{n} r_{mj} v_j = \lambda d_{mm} v_m$$

Taking the absolute value of both sides and substituting $|v_m| = 1$, produces:

$$|\lambda||d_{mm}| = \left| \sum_{j \neq m}^{n} r_{mj} v_j \right|$$

Applying the triangle inequality, we get:

$$|\lambda||d_{mm}| \leq \sum_{j \neq m}^{n} |r_{mj}||v_j|$$

Since $|v_j| \leq 1$ for all j, we can bound the right side:

$$|\lambda||d_{mm}| \leq \sum_{j \neq m}^{n} |r_{mj}|$$

By the hypothesis that A is strictly diagonally dominant, the absolute value of the diagonal entry strictly dominates the sum of the absolute values of the off-diagonal entries in its row:

$$\sum_{j\neq i}^{n} |a_{ij}| < |a_{ii}| \implies \sum_{j\neq m}^{n} |r_{mj}| < |d_{mm}|$$

Combining these inequalities gives:

$$|\lambda||d_{mm}| < |d_{mm}|$$

Since A is strictly diagonally dominant, its diagonal elements cannot be zero, so $|d_{mm}| > 0$. Dividing both sides by $|d_{mm}|$ yields:

$$|\lambda| < 1$$

Because λ was an arbitrary eigenvalue, it follows that $\rho(D^{-1}R) < 1$: this consequence is due to the fact that we have proven that all eigenvalues are less than one. We now rely on the general theorem on iterative methods:

Theorem A.7. *If the $n \times n$ matrix M has spectral radius $\rho(M) < 1$, and c is arbitrary, then, for any vector x_0, the iteration $x_{k+1} = Mx_k + c$ converges. In fact, there exists a unique x_* such that $\lim_{k\to\infty} x_k = x_*$ and $x_* = Mx_* + c$.*

Therefore, let $M = -D^{-1}(L+U)$ and $c = D^{-1}b$, we have:

$$x_{k+1} = Mx_k + c = -D^{-1}(L+U)x_k + D^{-1}b$$

This guarantees that the Jacobi method converges for any starting guess x_0 and any vector b.

Finally, since the linear system $Ax = b$ is guaranteed to have a unique solution for every possible b, fundamental linear algebra affirms that the matrix A must be invertible. Therefore, A is a non-singular matrix, completing the proof. $\square$